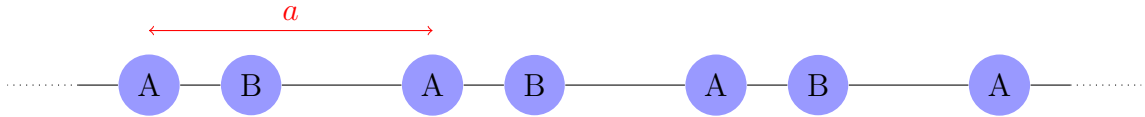


Homework 4

Translation symmetry

1. Consider the 1d chain with third neighbor springs, equal masses, and $N = \infty$. Assume that the springs between first neighbors, second neighbors, and third neighbors are γ_1 , γ_2 , and γ_3 , respectively.
 - (a) Construct the force constants $\Phi_0, \Phi_1, \Phi_2, \Phi_3, \Phi_{\bar{1}}, \Phi_{\bar{2}}, \Phi_{\bar{3}}$.
 - (b) Construct the Dynamical matrix for an arbitrary k vector.
 - (c) Plot the band structure for $\gamma_1 = \gamma$, $\gamma_2 = 4\gamma$, and $\gamma_3 = -\frac{1}{2}\gamma$. The y-axis should be in units of $\sqrt{\frac{\gamma}{m}}$.
 - (d) Now we will discover the notion of a soft mode. Repeat the previous plot, but now use $\gamma_3 = -\frac{11}{10}\gamma$. If you encounter imaginary numbers (hint), please plot them along the negative axis.
 - (e) Plot/sketch the energy of the system as a function of the displacement amplitude along $k = \frac{1}{4}$ and $k = \frac{1}{2}$ using the parameters $\gamma_1 = \gamma$, $\gamma_2 = 4\gamma$, and $\gamma_3 = -\frac{11}{10}\gamma$ (i.e. one line for each case). Please be sure to indicate the curvature in each case.
2. Consider the 1d chain with alternating nearest neighbor springs, with spring constants γ_1 and γ_2 .



The Hamiltonian can be written as follows:

$$H = \sum_{t=0}^{N-1} \left[\frac{(p_t^A)^2}{2m} + \frac{(p_t^B)^2}{2m} + \frac{1}{2}\gamma_1(x_t^B - x_t^A)^2 + \frac{1}{2}\gamma_2(x_{t+1}^A - x_t^B)^2 \right]$$

- (a) Construct the force constants.
- (b) Construct the dynamical matrix for an arbitrary k .
- (c) Assume that $\gamma_1 = \gamma(1 + \delta)$ and $\gamma_2 = \gamma(1 - \delta)$. Plot the band structure for $\delta = 0$ and for $\delta = 0.1$, with the y-axis in units of $\sqrt{\frac{\gamma}{m}}$.
- (d) Write an analytic equation for the band gap at the zone boundary (i.e. $k = \frac{1}{2}$) as a function of δ . Also, Taylor series expand in δ to obtain the leading order term.
- (e) Derive an analytic equation for the Eigenvectors at $k = \frac{1}{2}$ as a function of δ .
- (f) Sketch the Eigenvectors at $k = \frac{1}{2}$ for a small, positive value of δ .